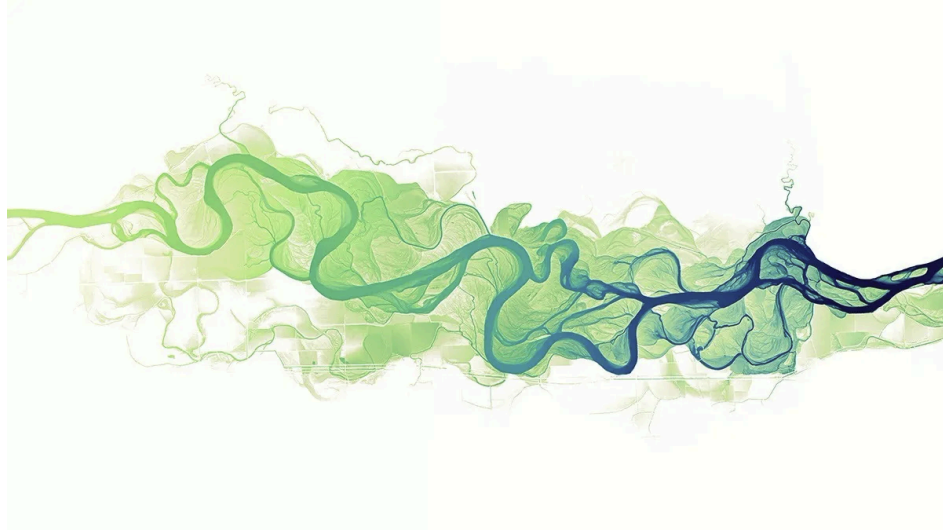


Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning



Bachelor Thesis

by
Florian Vollmer

Bachelors 2023

Submission date: 2026-07-02

Study Direction: Applied Digital Life Science

Supervisor:

Dr. Stefan Glüge

ZHAW Life Sciences und Facility Management, Wädenswil

Cross Corrector:

Prof. Dr. Robert Vorburger

ZHAW Life Sciences und Facility Management, Wädenswil

Imprint

Recommended Citation:

Florian Vollmer (2026). *Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning*

.

Keywords: Deep Learning, Cell Detection, Evaluation Metrics, Digital Lab and Production, Clinical Image Processing

Abstract

English Version

German Version

Acknowledgement

I would like to express my sincere gratitude to Dr. Stefan Glüge for setting up the project for this thesis and for his valuable support and guidance throughout the writing and compilation process of this thesis. My appreciation also goes to Robin Kosche for showing me around the USZ and introducing me to the subject.

Table of Contents

List of Abbreviations	6
1. Introduction	7
2. Methodology	8
2.1. Biology of Mast Cells and Systemic Mastocytosis	8
2.2. Technical	9
3. Results	24
3.1. Model Results and Statistical Evaluation	24
3.2. Quantitative and Qualitative Evaluation	27
3.3. Evaluation Visualization	29
3.4. Real World Usage	33
4. Discussion	35
4.1. Interpretation of Results	35
4.2. Clinical Relevance	35
4.3. Error Analysis	35
5. Conclusion	36
5.1. Summary of Key Findings	36
5.2. Limitations and Challenges	36
5.3. Future Research Directions	36
6. Declaration on AI Assistance	37
7. References	38
8. Lists	39
List of Figures	39

List of Tables	40
9. Attachments	41

List of Abbreviations

- **AI**: Artificial Intelligence
- **AP**: Average Precision
- **CNN**: Convolutional Neural Network
- **COCO**: Common Objects in Context
- **CV**: Cross Validation
- **CVAT**: Computer Vision Annotation Tool
- **DoE**: Design of Experiment
- **FN**: False Negative
- **FP**: False Positive
- **IoU**: Intersection over Union
- **LOPO**: Leave-One-Patient-Out
- **MC**: Mast Cells
- **mAP**: Mean Average Precision
- **NMS**: Non-Maximum Suppression
- **RF-DETR**: Roboflow Detection Transformer
- **SM**: Systemic Mastocytosis
- **TN**: True Negative
- **TP**: True Positive
- **USZ**: University Hospital Zurich (Universitätsspital Zürich)
- **WHO**: World Health Organization
- **YOLO**: You Only Look Once

1. Introduction

2. Methodology

2.1. Biology of Mast Cells and Systemic Mastocytosis

2.2. Technical

General Data Information

The dataset for this study consists of digitized bone marrow aspirate smear images provided by the University Hospital Zurich (USZ). All samples were collected from patients with confirmed or suspected systemic mastocytosis. Image tiles were extracted from whole-slide scans at a fixed resolution of 512x512 pixels, each displaying a sub-section of the whole smear. Two morphological classes were labeled for this study: Atypisch (atypical mast cells) and Normal (normal mast cells), corresponding directly to the cell types relevant to the WHO minor diagnostic criterion for SM.

The corpus grew across four acquisition batches spanning 53 patients. The initial dataset covered a single patient (P1) and served as the pilot for the annotation and training pipeline. A second batch extended the corpus to 15 patients (P1-P15), contributing 1,564 annotated images with 1,288 Atypisch and 391 Normal instances, reflecting a raw class ratio of approximately 3.3:1. A third batch added patients P16 through P53, bringing the full corpus to 53 patients. Because the training data derives almost entirely from confirmed SM patients, Atypisch is the dominant class throughout. At the P1-P8 scale the raw ratio was 14.4:1; by P1-P15 this had dropped to 3.3:1 due to the inclusion of patients P6, P8, and P9, which are rich in Normal annotations.

Data Acquisition and Annotation

The annotation pipeline was carried out by Robin Kosche at USZ using CVAT (Computer Vision Annotation Tool). Rather than annotating all 53 patients manually from scratch, an iterative model-assisted annotation strategy was used to make the labeling process tractable at the scale of tens of thousands of image tiles.

The first patient (P1) was annotated entirely by hand. Bounding boxes were drawn around each visible mast cell. This initial patient did only had Atypisch cells. This dataset was used to train the initial domain-specific detection model, DL_Modell_FV.pt, a YOLO11n checkpoint trained on P1 data.

For patients P2 through P15, CVAT's model inference integration was used with DL_Modell_FV.pt as the backend. The model generated candidate bounding boxes for each image tile, which the annotator then reviewed, accepted, corrected, or rejected. Starting from this batch, also normal cells were annotated. Tiles where the model produced detections on regions that contained no mast cell were recorded separately in per-patient Excel files. These confirmed false positives were later incorporated into training as a curated set of hard-negative examples.

For patients P16 through P53, the same workflow was repeated using a second-generation model (DL_Modell_FV_v2.pt), trained on the P1-P15 corpus, as the inference backend. It is worth noting that a subset of patients in this batch contained no annotatable mast cells at all: the model produced candidate detections, but all were rejected by the annotator as false positives. These patients contributed no positive labels to the corpus but supplied a large number of confirmed hard negatives. The annotations from the full P1-P53 corpus were then used to train a third-generation model, DL_Modell_FV_v3.pt, completing the bootstrapped annotation cycle at the current project scope.

This iterative approach, where each annotation round produces a labeled dataset used to train a better model for the next round, allowed the corpus to grow from one patient to 53 without requiring fully manual annotation of all 53 cases. The confirmed false positives collected at each round provided a particularly valuable training signal: these are the specific tiles that the model of that generation found difficult to reject, making them targeted hard negatives for the next training cycle.

This bootstrapped annotation strategy is not documented for mast cell detection in the existing literature. The systematic review by Ghete et al. [1], covering AI-based bone marrow cell classification from 2019 to 2024, found that the majority of published studies applied their models to pre-existing datasets rather than developing annotation pipelines alongside the model. In this project, annotation and model development were tightly coupled by design: the clinical partner’s annotation capacity was directly extended by each generation of model, and the confirmed false positives produced during annotation became training signal for the next generation. This coupling is what made a 53-patient corpus feasible within the scope of a single thesis.

The final corpus assembled across all three annotation rounds is summarized in Table 1.

Table 1: Full corpus summary across P1-P53.

Category	Count
Total annotated images	1,869
Atypisch cell labels	1,473
Normal cell labels	511
Confirmed FP/Background images	4,940

Data Preprocessing

Two distinct preprocessing approaches were applied across the experimental timeline, with a significant revision midway through the project.

Pre-applied augmentation (Phase 1 and early Phase 2). For the initial experiments, a preprocessing pipeline was implemented using the `albumations` library. Each annotated image was transformed once before training with random flips, brightness and contrast adjustments, and affine rotations of up to 15 degrees. The augmented copies were saved to disk and used as fixed training input. This approach proved ineffective: because transforms were applied once and stored, every training epoch saw the same augmented copies. The augmented images extended the dataset by a fixed factor but provided no additional regularization benefit after the first epoch, which is the only epoch where the transforms are novel to the model. This limitation was identified during Phase 2 and motivated the switch to on-the-fly augmentation.

On-the-fly augmentation (Phase 3 onward). Starting from Phase 3, augmentation was moved into the YOLO training loop by enabling `augment=True` in the `model.train()` call. This applies a fresh random transform to each image at every epoch, meaning oversampled copies in the training file list present different pixel content to the model across epochs. Two augmentation parameters required explicit configuration for this domain. The mosaic augmentation was disabled (`mosaic=0.0`): YOLO’s mosaic mode composites four images into a single training sample, which is appropriate for multi-object scene datasets like COCO but is counterproductive for 512x512

bone marrow tiles, where the spatial relationship between a candidate cell and its immediate local context is part of the morphological signal the model needs to learn. Vertical flip probability was set to 0.5 (the YOLO default is 0), consistent with the biological reality that bone marrow mast cells have no canonical vertical orientation and that the model should not learn a vertical bias from the training data.

The rotation augmentation parameter (degrees) was treated as a tunable hyperparameter during Phase 3. A systematic tinkering matrix identified degrees=5 as optimal at the Phase 3 corpus scale, providing approximately 4-5 percentage points of improvement in Normal-class recall over degrees=0. The effect is biologically justified: mast cells present in any orientation in aspirate smears, and moderate rotation augmentation regularizes the model without distorting morphological cues that distinguish Atypisch from Normal.

Model Selection and Architecture

YOLO11 (Ultralytics) was selected as the object detection framework. The architecture performs detection and classification in a single forward pass, making it computationally efficient for integration into clinical workflows where inference throughput matters. The Ultralytics Python API provides a consistent interface for training, fine-tuning, evaluation, and inference, and supports checkpoint initialization from custom pretrained weights, which was central to the transfer learning strategy described below. Comparable studies applying YOLO-family models to cell detection in hematological microscopy report mAP50 values in the range of 77-94% on well-represented cell types [2] [3], placing the architecture in the performance range expected for this domain.

Among the available YOLO11 model sizes, the nano variant (yolo11n, 2.6M parameters) was selected. The selection was validated experimentally: a test using the large variant (yolo11l, 43M parameters) on the P1-P8 corpus under identical training conditions produced $R_{\text{Normal}}=0.695$ compared to $R_{\text{Normal}}=0.840$ for yolo11n. The larger model's classification head, which is randomly re-initialized when training with a different number of output classes than the pretrained checkpoint, has proportionally more free parameters to adapt from scratch. With only 170-220 training images per LOPO (Leave-One-Patient-Out) fold, these parameters tend to overfit to the majority class rather than learn generalizable features for the rare Normal class. The corpus size at this stage was insufficient for a 43M-parameter model; yolo11n was confirmed as the appropriate architecture for the available data.

Throughout the project, all training runs started from the generic COCO-pretrained yolo11n.pt checkpoint. The single exception was a set of experiments during Phase 2 in which DL_Modell_FV.pt, a YOLO11n model previously trained on the P1 single-patient dataset (nc=1, one class), was used as the starting point for continued training on the two-class P2 dataset. The reasoning was that domain-specific pretraining might give the model a better starting point for bone marrow aspirate images compared to COCO features. In practice, this approach introduced a structural complication: because DL_Modell_FV.pt was trained with a single output class and the P2 training required two classes (Atypisch and Normal), the Detect head's classification branch (cv3, the final 1x1 convolution layer) had to be discarded and randomly re-initialized. The bounding-box regression branch (cv2) transferred cleanly, but the randomly initialized cv3 layer produced noisy classification gradients in early training epochs. At the default learning rate of $lr=0.01$, these gradients were large enough to partially overwrite the pretrained neck features,

causing systematic underperformance rather than the expected improvement. The full account of this failure and its resolution is described in the training strategy section. After Phase 2, the project returned to yolo11n.pt as the standard base for all subsequent training.

Model Training and Finetuning Strategy

The training strategy developed across five phases, each responding to specific failures and insights from the previous phase. The following account describes the major decision points rather than every individual run; a complete run table is provided in the Results section. The phase structure corresponds to successive expansions of the training corpus and methodology: Phase 1 covered a single pilot patient, Phase 2 moved to a two-class annotated dataset and explored data composition, Phase 3 introduced multi-patient LOPO cross-validation and on-the-fly augmentation on an 8-patient corpus, Phase 4 expanded to 15 patients, and Phase 5 targets the full 53-patient corpus. Individual experiment runs within each phase are identified by a label combining the phase number and a letter (e.g., P2-C, P3-A), which matches the experiment log maintained throughout the project.

Phase 1: Single-patient proof of concept and hyperparameter search.

Phase 1 covered a sequence of experiments on the P1 dataset (428 images, single patient, Atypisch class only) using 5-fold stratified cross-validation and the pre-applied augmentations augmentation pipeline described above. The dataset consisted of annotated positive tiles alongside hand-curated negative tiles (confirmed empty regions), all at 512x512 pixels.

The initial runs used yolo11n.pt (COCO pretrained). To test whether a larger model would learn better features for this domain, yolo11l (43M parameters) was then substituted and run in three separate configurations, including extended epoch budgets (200 epochs) and added augmentation. In each case, the large model did not outperform yolo11n on this single-patient dataset, and the project returned to yolo11n for all further Phase 1 work. This finding prefigured the same result later confirmed empirically at larger scale in Phase 3 (P3-E), where yolo11l again underperformed yolo11n on a multi-patient corpus.

With yolo11n established as the appropriate architecture, hyperparameter tuning was run using YOLO's built-in `model.tune()` function across three increasingly intensive rounds: 30 iterations x 50 epochs, 300 iterations x 50 epochs, and finally 300 iterations x 100 epochs. The output of the final tuning round was a `best_hyperparameters.yaml` file with optimized values including `lr0=0.0062`, `momentum=0.791`, `mosaic=0.916`, and `degrees=0.0`. A training run applying these tuned hyperparameters completed Phase 1. The per-fold results are reported in the Results section.

The `best_hyperparameters.yaml` produced by the Phase 1 tuner is significant beyond Phase 1: it was later applied to fine-tuning on P2 data (P2-E) and caused catastrophic forgetting, collapsing performance across all five folds. The failure was later attributed to a recipe mismatch: the tuner had optimised on a from-scratch yolo11n training run, and its output (particularly the high mosaic value and `degrees=0.0`) was calibrated for scratch training from COCO, not for fine-tuning in this clinical domain. The Phase 1 yaml was permanently retired from the fine-tuning pipeline as a result.

Following the 5-fold training, an inference run was conducted on the full first-patient slide image set, approximately 20,000 tiles, excluding all training images by filename. The resulting confidence

distribution provided an early look at the model’s behavior at deployment scale and informed the direction of subsequent annotation and training work.

Phase 2: The domain-specific checkpoint experiment and the learning rate problem.

Phase 2 began with two runs using the standard yolo11n.pt base on the annotated P2 dataset (P2-A, P2-B), but both were corrupted by a concurrent SLURM job collision described below. The third run (P2-C) introduced DL_Modell_FV.pt as the starting point, testing whether the P1-trained domain-specific model would give the detector a better foundation than COCO-pretrained weights. The result was a characteristic failure pattern: high precision combined with very low recall, the signature of a model that makes few predictions and misses most cells. The cause was the interaction between the randomly re-initialized cv3 head and the default learning rate of $lr0=0.01$: the noisy gradients from the freshly initialized classification branch were large enough to partially overwrite the pretrained neck features during early epochs, leaving the model worse than yolo11n.pt would have been from scratch.

Reducing the learning rate to $lr0=0.001$ recovered the performance lost to this overwriting effect. Freezing the first ten backbone layers ($freeze=10$) was introduced alongside the learning rate change to prevent the small dataset from further disrupting features that COCO pretraining had established. This combination ($lr0=0.001$, $freeze=10$) defined the reference configuration for the remainder of Phase 2. However, the conclusion drawn from the full Phase 2 experiment set was that using DL_Modell_FV.pt as a starting point added complexity without a measurable benefit over yolo11n.pt, and all subsequent phases returned to the standard COCO-pretrained base.

Before expanding to the full P1-P15 corpus that would produce the second model version, a 3×4 factorial design-of-experiment (DoE) grid was run to identify the best data composition strategy at Phase 2 scale. The goal was practical: the hyperparameter and data configuration choices made here would carry forward into the P1-P15 training, and a better second model version meant fewer manual corrections for the annotation partner at USZ when working through patients P16-P53. The grid varied confirmed FP negative oversampling (1x, 2x, 3x) and random background tile ratio (0, 1, 2, 3) across twelve configurations. The mAP50 spread across all twelve DoE configurations was smaller than the cross-fold standard deviation within any single run. Data composition manipulations at single-patient scale could not produce statistically interpretable performance differences; the limiting factor was data scarcity rather than class balance. Background tiles uniformly hurt performance at every FP oversampling level, and this finding was consistent enough to permanently set the random background ratio to zero in all subsequent runs. The DoE finding reframed the problem: improving recall required more patients and better augmentation, not a different ratio of existing data categories. This result has broader relevance for rare-event medical imaging tasks where the class imbalance is not accidental but structural: when a dataset is collected from disease-positive patients, the minority class is underrepresented because the disease itself is rare in that morphological form, not because of sampling bias. In that regime, reweighting or resampling the existing data cannot substitute for collecting more biologically diverse examples. The systematic nature of the DoE grid makes this conclusion more than a project observation; it is a controlled negative result that points practitioners toward corpus expansion as the primary lever for rare-class recall improvement.

Phase 2 also exposed an infrastructure failure specific to the HPC environment. Submitting two SLURM jobs concurrently caused both jobs to share the same temporary workspace directories, because directory paths were derived from fold indices without job-level namespacing. Concurrent writes to shared directories corrupted three of five folds in each job, detectable by fold results

that were bit-for-bit identical between the two affected runs. The fix applied `$SLURM_JOB_ID` namespacing to all temporary directories and derived the output project directory name from the job name set at sbatch submission time, making concurrent jobs structurally independent.

Phase 3: Multi-patient LOPO and on-the-fly augmentation.

Phase 3 restructured the training and evaluation design around three simultaneous changes: expansion to 8 patients, adoption of Leave-One-Patient-Out cross-validation, and the switch to on-the-fly augmentation.

The LOPO design was motivated by a systematic bias in the Phase 1 and Phase 2 stratified k-fold protocol. When images from the same patient appear in both training and validation splits, the model can exploit patient-specific staining characteristics and morphological style to improve validation metrics without learning features that generalize to new patients. LOPO eliminates this by holding out all images from a single patient per fold, making each fold an evaluation of genuine cross-patient generalization.

The combined effect of these three changes was substantial. The Phase 3 baseline run (P3-A: 8 patients, yolo11n.pt, augment=True, mosaic=0.0, cls=1.0) achieved the strongest results seen to that point in the project, with cross-fold standard deviation markedly lower than the Phase 2 DoE despite the harder evaluation protocol: the more diverse training corpus more than compensated for the stricter holdout design. Full results are reported in the Results section.

With the same goal of finding the best possible training recipe before the P1-P15 training that would produce the second model version, a tinkering matrix of 14 parameter combinations explored rotation degrees (0, 5, 10), distribution focal loss weight dfl (1.5, 2.0), backbone freeze depth (0, 10), and label smoothing (0.0, 0.1). The production recipe confirmed here would be directly applied to the P1-P15 training: a higher-quality second model version meant the annotation partner at USZ would need fewer manual corrections when processing patients P16-P53. Rotation degrees was the dominant factor: degrees=5 consistently improved R_Normal over degrees=0 across all freeze and dfl combinations. Mast cells present in all orientations in aspirate smears, and moderate rotation augmentation directly targets this. The dfl parameter showed a consistent directional effect: dfl=2.0 regressed R_Normal relative to dfl=1.5 across all other parameter combinations, suggesting that increasing localisation loss weight tilts model capacity away from the classification discrimination needed for the rare Normal class. Freeze depth confirmed the Phase 2 finding: freeze=10 outperformed freeze=0 in every recall-relevant comparison. Label smoothing proved inert in this Ultralytics version, likely because the implementation does not propagate the parameter through the classification loss path.

The production recipe confirmed by the tinkering matrix was: degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0, augment=True, mosaic=0.0, flipud=0.5.

An automated hyperparameter search using YOLO's built-in `model.tune()` was run on the production recipe (30 iterations, 50 epochs, patient P4 as holdout, mAP50-95 as fitness). The tuner converged on degrees=0.0 and dfl=2.47, both directions that the tinkering matrix had identified as harmful to R_Normal. The explanation is structural: a single-fold tuner optimizing mAP50-95 on a P4 holdout (an Atypisch-dominant patient) maximizes Atypisch-class performance at the expense of Normal-class generalization. Single-fold automated tuning is blind to cross-patient per-class recall balance. The production recipe from the tinkering matrix was retained.

A model-size comparison test (P3-E) substituted yolo11l (43M parameters) for yolo11n (2.6M parameters) under identical conditions on the P1-P8 corpus. R_Normal dropped to the lowest

value measured across the entire project. The larger model’s randomly re-initialized classification head has more free parameters, and at approximately 170-220 training images per fold the model memorized Atypisch patterns rather than learning generalizable Normal features. This result confirmed yolo11n as the appropriate model size for the available corpus.

The production recipe confirmed by Phase 3 was carried forward and applied to train on the full P1-P15 corpus, producing the second model version (DL_Modell_FV_v2.pt) that the annotation partner at USZ used for patients P16-P53.

Phase 4: Corpus expansion to 15 patients.

Phase 4 expanded the training corpus to 15 patients (1,564 images, 1,288 Atypisch, 391 Normal, 2,464 Background images incl. FP; raw ratio 3.3:1, effective ratio 1.5:1 after automatic per-patient oversampling). The most significant addition was patient P9, which contributed 195 Normal annotations, more than doubling the total Normal count relative to the P1-P8 corpus. A 15-fold LOPO protocol was applied.

The first Phase 4 run (P4-A) was configured with degrees=0 due to an omitted configuration variable; the second run (P4-B) applied the full production recipe including degrees=5. The delta between the two runs was within the cross-fold standard deviation on all metrics, indicating that the benefit of degrees=5 observed at P1-P8 scale either diminishes with a more diverse 15-patient corpus or is within measurement noise at this scale. A freeze ablation run (P4-C, freeze=0) produced results essentially equivalent to P4-B across all metrics, confirming that backbone freezing remains appropriate and the P3-B recipe transfers to the larger corpus without re-tuning. Full per-fold results for P4-A, P4-B, and P4-C are reported in the Results section.

Phase 5: Full corpus (P1-P53). [Results pending]

The Phase 5 training run (P5-A) was submitted to the ZHAW HPC cluster during the final stages of this thesis. The run uses the confirmed production recipe (degrees=5, dfl=1.5, freeze=10, lr0=0.001, cls=1.0), fully automatic per-patient oversampling targeting a 2:1 Atypisch:Normal ratio, and a LOPO design across all annotated patients after filtering out patients with no mast cell labels. Results were not available at the time of writing and will be incorporated in the final version of this thesis.

Model Testing

Performance estimates in this thesis are reported under two different evaluation protocols depending on the phase. Phase 1 and Phase 2 used stratified k-fold cross-validation (5 folds), where images were split randomly across folds with class stratification. From Phase 3 onward, the evaluation protocol was changed to LOPO, where all images from one patient are withheld from training entirely and used exclusively for evaluation. The switch was motivated by a systematic bias identified in the Phase 1 and Phase 2 results: stratified k-fold splits on small medical imaging datasets allow patient-level texture and staining style to leak from training into validation, artificially improving validation metrics without reflecting true cross-patient generalization. LOPO results and stratified k-fold results are therefore not directly comparable and are reported separately in the Results section.

Choosing LOPO over stratified k-fold directly addresses a gap identified in the broader literature. Ghete et al. [1] found that performance metrics across published bone marrow cell classification studies are not directly comparable in part because most studies evaluate on subsets of their own

data, with few reporting results on patient-held-out splits and fewer still validating externally. LOPO is the strongest within-dataset approximation of external validation available without access to a second institution’s data: each fold evaluates the model on a patient it has never seen in any form, under the same conditions a deployed system would face in routine clinical use. The cross-fold standard deviation reported in this thesis therefore reflects genuine inter-patient biological variability rather than statistical noise from random sampling, making the performance estimates more conservative and more informative than single-split or random k-fold results typically reported in the field.

Each fold produced a training split, a validation split used only for early stopping during training, and a test split consisting of the held-out patient’s images. The metrics reported throughout this thesis refer to the test split. Across folds, the mean and standard deviation of each metric were computed. The cross-fold standard deviation reflects genuine inter-patient biological variability; patients with very small annotated sets (for example, P13 with 7 images) produce test metrics with extreme sensitivity, where a single missed cell changes recall by more than 14 percentage points.

One measurement limitation was identified in the per-class recall extraction. The implementation indexed Ultralytics’ per-class result arrays by position rather than by class label index. When one class was absent from both ground truth and predictions in a given fold, Ultralytics returned a shorter result array and positional indexing misassigned recall values. This affected primarily the P13 holdout fold in every run, where the reported `R_Normal` was NaN while the true Normal recall appeared in the `R_Atypisch` position. Folds affected by this issue are excluded from per-class mean calculations throughout this thesis, and the directions of the per-class trends are verified against folds where the per-class counts are large enough to confirm correct assignment.

Evaluation Metrics

Model performance was assessed using standard metrics from object detection. Each detection outcome is first classified into one of four categories based on whether the model’s predicted bounding box overlaps sufficiently with a ground-truth annotation (at a minimum Intersection over Union of 0.50):

- **True Positive (TP):** the model correctly detected a mast cell and the predicted box sufficiently overlapped the ground-truth annotation.
- **True Negative (TN):** the model correctly produced no detection in a tile containing no annotated mast cell.
- **False Positive (FP):** the model predicted a mast cell where the ground truth showed none (a false alarm).
- **False Negative (FN):** the model failed to detect a mast cell that was present in the ground truth (a missed cell).

From these, the following performance indicators were computed [PA2_FV]:

Precision measures the proportion of predicted detections that were correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures the proportion of ground-truth mast cells that the model successfully detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

mAP50 (Mean Average Precision at IoU 0.50) is the mean of the per-class Average Precision computed at a bounding-box overlap threshold of 50%. AP is the area under the precision-recall curve; mAP50 averages this across all classes:

$$\text{mAP50} = \frac{1}{N} \sum_{i=1}^N AP_{i@50}$$

mAP50-95 extends mAP50 by computing AP (Average Precision) at ten evenly spaced IoU thresholds from 0.50 to 0.95 (`torch.linspace(0.5, 0.95, 10)`) and averaging the result across thresholds and classes [4]. It penalises imprecise bounding box localisation more heavily than mAP50.

Recall is the primary metric for this project. A missed mast cell (false negative) causes the Atypisch fraction to be underestimated, potentially placing a patient below the WHO 25% diagnostic threshold and delaying diagnosis of SM. A false positive is less consequential for two reasons: the clinical workflow includes a manual review step where false alarms can be dismissed, and the diagnostic impact of a false positive depends on how far the true ratio is from the 25% threshold. In a patient with a strongly atypical cell population, a moderate false positive rate does not change the clinical conclusion because the Atypisch fraction remains clearly above the threshold. False positives become clinically relevant primarily in borderline cases where the ratio is close to 25%, and it is precisely in those cases that the manual review step is most likely to be exercised.

Two per-class recall metrics are tracked separately throughout the experiments: `R_Atypisch` (recall for the Atypisch class) and `R_Normal` (recall for the Normal class). Both are clinically necessary because the WHO criterion depends on the ratio of Atypisch to total detected mast cells. Undercounting Normal cells inflates this ratio, potentially triggering a false-positive SM classification in borderline patients. Tracking both classes independently, rather than reporting a single aggregate recall, is a direct consequence of aligning the evaluation to the diagnostic question rather than to standard object detection benchmarking practice. Published studies in adjacent domains, such as general white blood cell subtype classification, typically optimize for aggregate mAP or F1 across all classes [3]; this thesis treats `R_Normal` as a primary metric in its own right because the WHO threshold makes the Normal count as diagnostically load-bearing as the Atypisch count. `R_Normal` received particular experimental attention because Normal is the minority class in the training corpus, consistently exhibiting lower recall and higher cross-fold variability than `R_Atypisch`.

mAP50 is used as the standard benchmark for run-to-run comparison across the experimental timeline, providing a single-number summary that captures both detection and classification quality. mAP50-95 is reported where available but assigned less interpretive weight; localisation precision at high IoU thresholds is clinically less relevant than detection recall for this task. Precision is monitored throughout but treated as a secondary metric.

UI Creation

Gradio Prototype

An initial web-based interface was developed using Gradio to support early-stage result visualization and model evaluation during development. The prototype provided two functional tabs. The Analysis tab ran batch inference on image directories and displayed tiles with YOLO detection overlays (Figure 1), producing labeled output images. The Research tab supported ground-truth versus prediction comparison (Figure 2, Figure 3), applying a lower confidence threshold to maximize recall sensitivity during evaluation. A statistics dashboard (Figure 4) aggregated per-class detection counts and displayed the Atypisch fraction relative to the WHO 25% threshold. The Gradio prototype served its purpose as a rapid development tool and was later superseded by the desktop application.

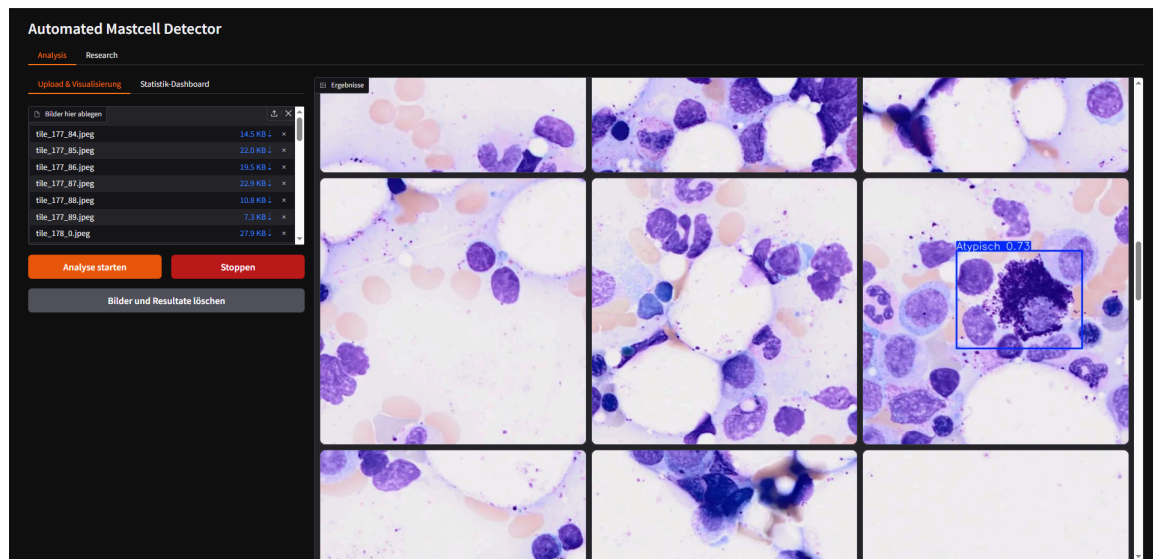


Figure 1: Analysis tab of the Gradio prototype showing uploaded image tiles with YOLO detection overlays.

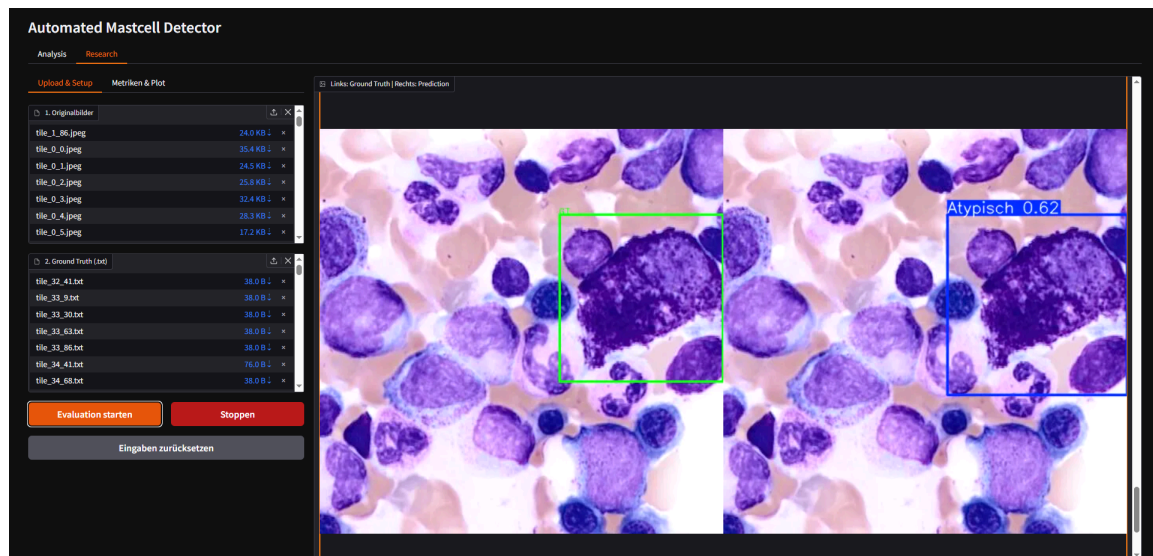


Figure 2: Research tab showing ground-truth annotations alongside model predictions for direct visual comparison.

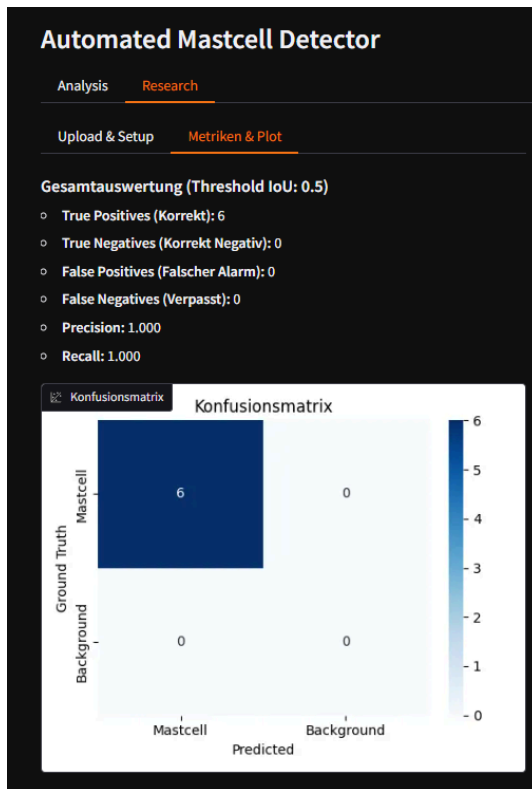


Figure 3: Research tab metrics output: precision-recall curves and performance plots generated from the ground-truth evaluation run.

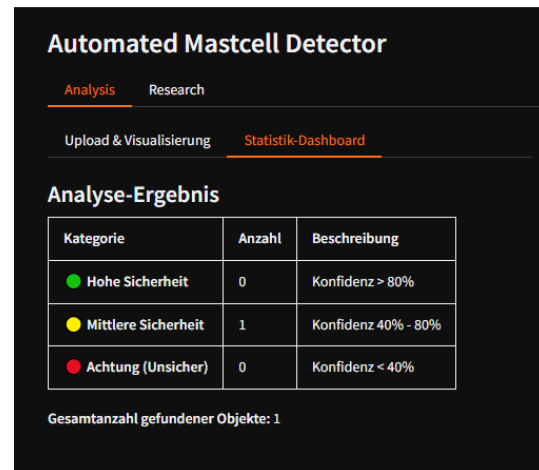


Figure 4: Statistics dashboard of the Gradio prototype showing per-class detection counts and the Atypisch fraction relative to the WHO 25% diagnostic threshold.

Desktop Application

As the requirements for clinical deployment became clearer, it became apparent that a browser-based interface was not adequate for the actual usage context at USZ. A full patient slide scan produces approximately 20,000 image tiles, and a Gradio-based interface was not designed for streaming inference at that scale. Cinderella, the cell classification software already in use at USZ, was considered as an integration target, but embedding the pipeline into an existing hospital codebase within the thesis timeline was not feasible: the time required to understand a production clinical codebase combined with the security validation and change management processes required for hospital software integration extended well beyond the project scope.

A native desktop application (MastCellDetector) was developed as the primary clinical deliverable. The application was built with PyQt and packaged using PyInstaller into a self-contained executable that requires no Python environment on the target machine. While the primary deployment target is Windows workstations at USZ, PyInstaller equally supports Linux builds, making the same application distributable on Linux servers or HPC systems.

The intended workflow for a USZ analyst proceeds as follows. The analyst opens a local folder containing a patient's image tiles. The sidebar provides controls for confidence threshold, IoU non-maximum suppression threshold, and hardware selection (GPU with automatic VRAM-based batch size recommendation, or CPU fallback). Running inference streams YOLO-format label files to disk alongside each image in real time, so results are preserved if the run is interrupted.

The gallery view (Figure 5) displays all images with filter tabs for All images, With detections (Figure 6), No detection, Low confidence, Atypisch only, and Normal only. This allows a haematologist to navigate directly to the cases most likely to need manual review rather than scrolling through tens of thousands of tiles.

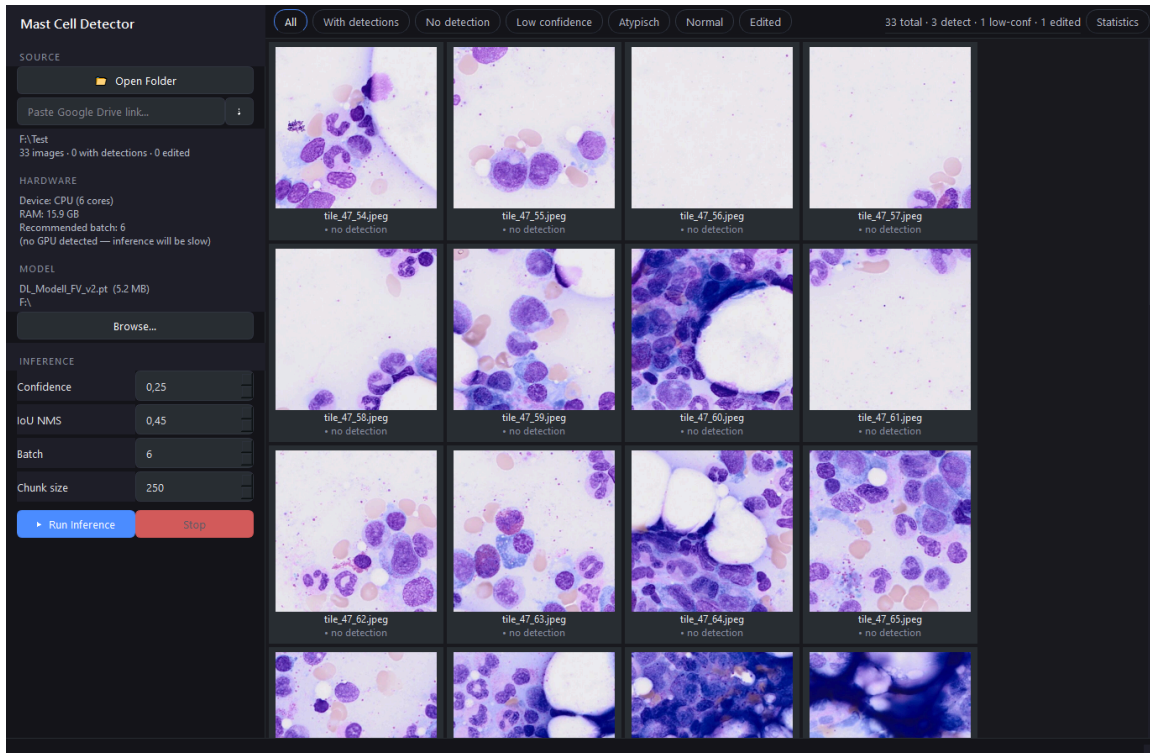


Figure 5: Gallery view of the desktop application showing all image tiles for a patient folder with sidebar controls for inference configuration.

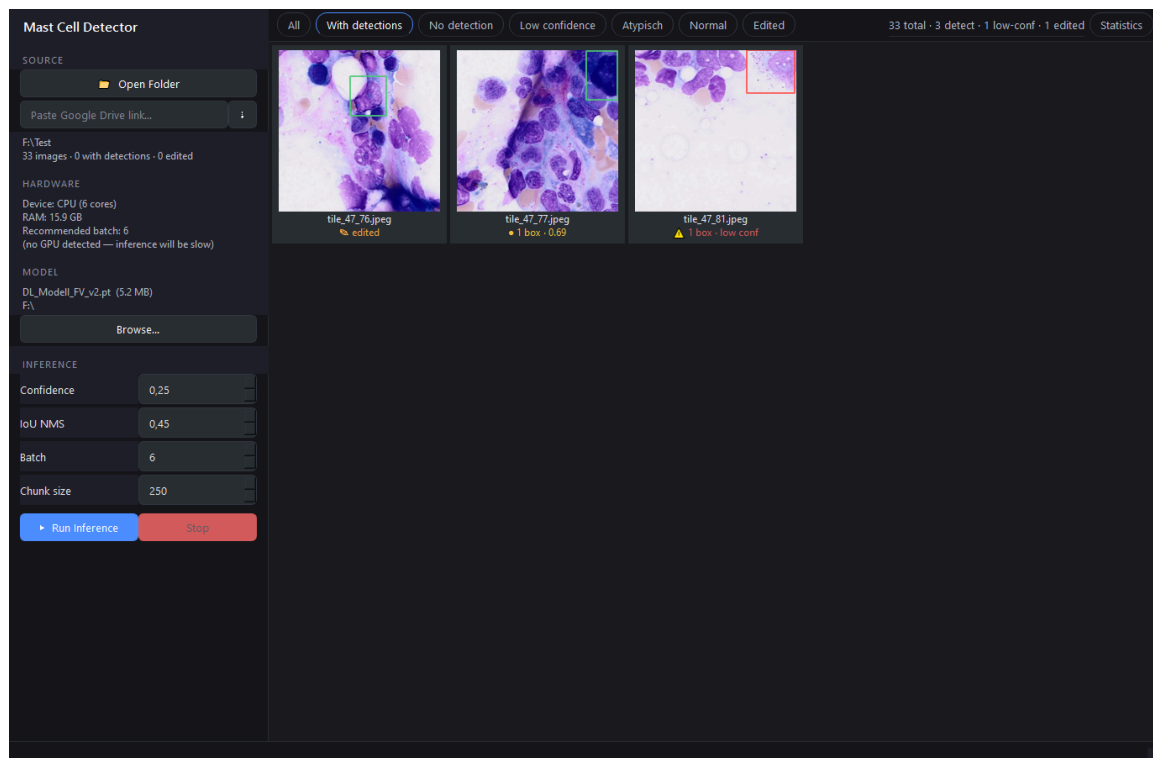


Figure 6: Gallery filtered to tiles with detections, allowing rapid navigation to images where the model found mast cells.

An annotation editor is embedded in the gallery. Clicking any thumbnail opens the image with interactive bounding boxes (Figure 7): boxes can be repositioned by dragging, resized by dragging corner handles, and deleted with the Delete key. New boxes are drawn by pressing A and clicking and dragging a region, with class selected from a dropdown. Changes are saved to the corresponding label file with Ctrl+S or automatically on navigation to the next image. When a box is deleted as a false positive, an empty label file is written, marking the tile as a confirmed negative for any future retraining cycle. Edited files are protected from being overwritten by subsequent inference runs.

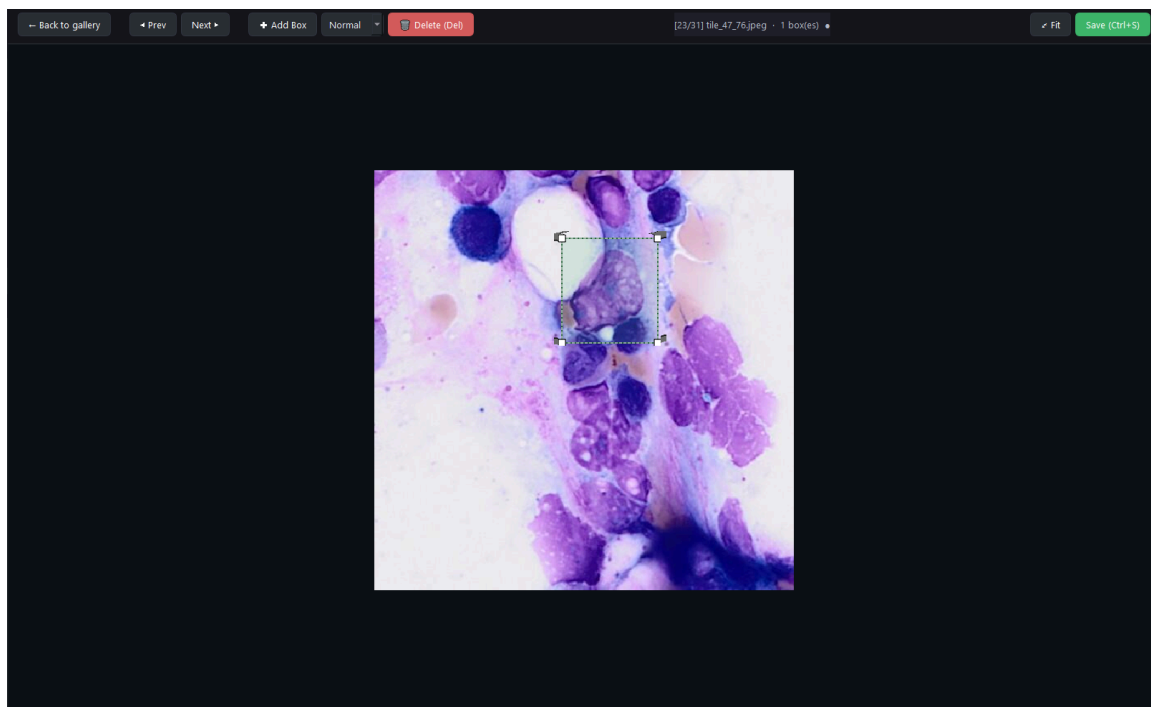


Figure 7: Annotation editor with interactive bounding boxes; boxes can be added, moved, resized, or deleted directly in the application without external tooling.

The Statistics tab (Figure 8) aggregates per-image detection counts into a patient-level summary that displays the total detected mast cells, the Atypisch fraction as a percentage, and a visual marker relative to the WHO 25% diagnostic threshold. This summary can be exported for clinical documentation or further analysis.

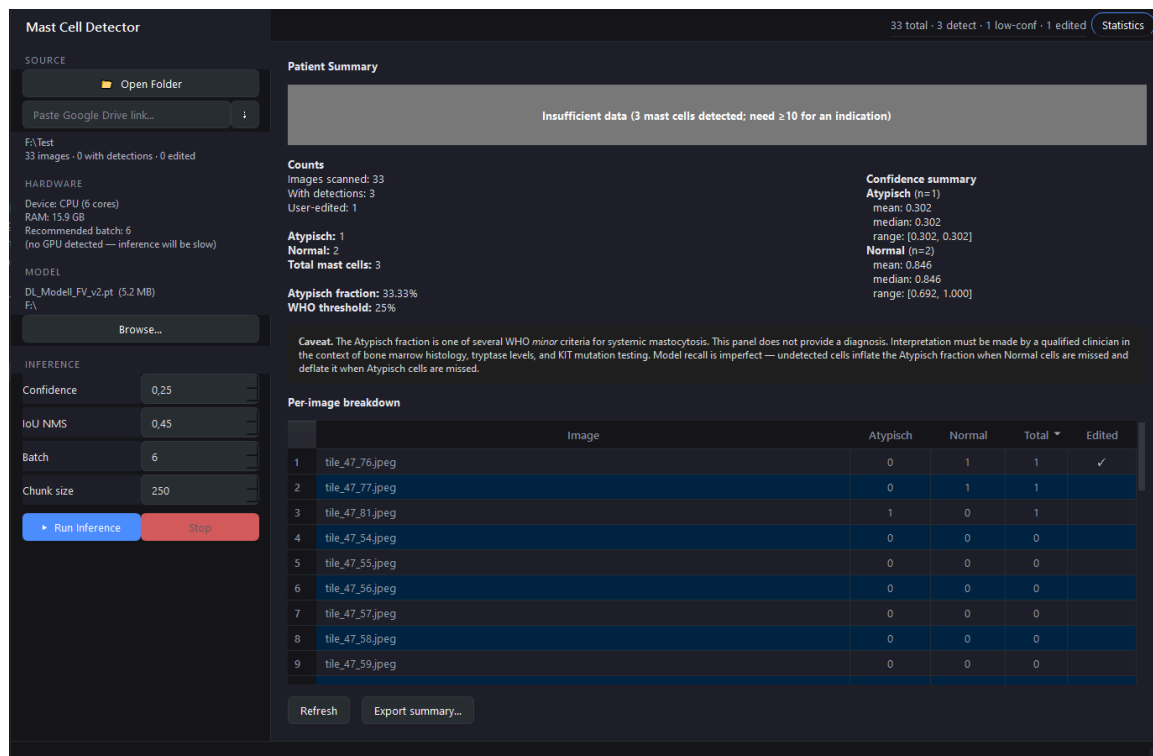


Figure 8: Statistics tab aggregating per-image detections into a patient-level summary with the Atypisch fraction displayed against the WHO 25% diagnostic threshold.

Implementation in USZ Infrastructure

The primary deliverable of this project is the MastCellDetector desktop application, distributed as a self-contained Windows directory. The application operates entirely offline after installation, which satisfies a practical requirement for deployment in hospital network environments where internet access from clinical workstations is restricted or prohibited.

The model shipped with the final version of the desktop application will be DL_Model_FV_v3.pt, trained on the complete P1-P53 corpus using the production recipe established across Phases 3 and 4. This model was not yet available at the time of writing.

The annotation editing and label export functionality in the application also addresses a secondary deliverable: the confirmed false positives and corrected labels produced during clinical use feed directly back into the training pipeline as the next generation of hard-negative examples, continuing the iterative annotation strategy established from Phase 1 onward.

The Linux build capability opens a second deployment path beyond the clinical workstation. A Linux build of MastCellDetector can be deployed on a computational server or HPC node at USZ, allowing the tool to be integrated directly into the institution’s scan-to-inference-to-evaluation pipeline: whole-slide images are scanned, tiles are passed through the model at server-side GPU throughput, and the resulting label files and statistics export are produced without requiring a Windows workstation in the loop. This makes the same application suitable for both interactive clinical review and automated batch processing, depending on the deployment context. The source code of the application is publicly available [5].

3. Results

3.1. Model Results and Statistical Evaluation

Performance metrics for Phase 3 and Phase 4 runs are reported from the test split of each LOPO fold. Phase 1 and Phase 2 results derive from held-out validation folds under 5-fold stratified cross-validation and are not directly comparable to Phase 3–4 LOPO estimates, for the reasons described in the Model Testing section. All values quoted in this section refer to cross-fold means unless stated otherwise.

Table 2 summarizes the key milestone runs across all phases. The complete experiment record, including corrupted runs and all 11 DoE grid cells, is maintained in `hpc/experiment_log.md`. `R_Normal` for P3-A is reported as the mean over the four Normal-rich holdout folds only (P5, P6, P7, P8, each with at least 6 Normal instances); folds with fewer than 3 Normal instances in the test set (P1: 2, P3: 4) are excluded from that average because a single missed cell produces metric swings of 50 and 25 percentage points respectively, making those values dominated by small-sample noise rather than model behaviour.

Table 2: Milestone run progression across all experimental phases. Phase 1–2 values are from stratified k-fold validation splits; Phase 3–4 values are from LOPO test splits. ★ = production recipe confirmed by P3-B tinkering matrix. ★★ = best complete run, cited as the primary performance reference throughout this thesis.

Run	Phase	CV	Base model	mAP50 (mean±std)	Recall	R_Atypisch	R_Normal	Verdict
P1-base-line	1	5-fold strat.	yolo11n	0.800 ± 0.129	0.719	—	—	Proof of concept
P2-C	2	5-DL_Modell_FV fold strat.		0.469 ± 0.031	0.397	—	—	LR mismatch
P2-H	2	5-DL_Modell_FV fold strat.		0.604 ± 0.194	0.537	—	—	Phase 2 reference
DoE ceiling	2	5-DL_Modell_FV fold strat.		0.604	0.638	—	—	fp=1, bgr=0
P3-A	3	8-fold LOPO	yolo11n	0.844 ± 0.137	0.840	0.884†	0.861‡	Major improvement
P3-B ★	3	8-fold LOPO	yolo11n	0.864	0.866	0.892	0.840	Production recipe
P3-E	3	8-fold LOPO	yolo11l (43 M)	0.855	0.785	—	0.695	Model-size failure
P4-A	4	15-fold LOPO	yolo11n	0.903 ± 0.100	0.904	0.913	0.890	Major improvement
P4-B ★★	4	15-fold LOPO	yolo11n	0.887	0.895	0.900	0.882	Best complete
P4-C	4	15-fold LOPO	yolo11n	0.883	0.893	0.896	0.882	Freeze ablation null
P5-A	5	LOPO (P1–P53)	yolo11n	TBD	TBD	TBD	TBD	Pending

†P3-A R_Atypisch estimated from the P3-B tinkering matrix default cell, which reproduces P3-A exactly with identical configuration. ‡P3-A R_Normal computed as mean over four Normal-rich holdout folds (P5, P6, P7, P8).

Phase 1. The P1-baseline achieved validation $\text{mAP50} = 0.800 \pm 0.129$ and $\text{recall} = 0.719$, establishing that detection on 512×512 bone marrow aspirate tiles was feasible from COCO-pretrained weights. The high cross-fold variance ($\text{std} = 0.129$) reflected the instability of random stratified splits on a single 428-image patient: Fold 1 reached $\text{mAP50} = 0.952$ while Fold 5 collapsed at epoch 29 ($\text{recall} = 0.432$). No per-class breakdown was available because P1 contained only Atypisch annotations.

Phase 2. The 3×4 factorial DoE grid (11 cells, FP oversampling $\in \{1\times, 2\times, 3\times\}$ crossed with background ratio $\in \{0, 1, 2, 3\}$) produced a total mAP50 range of 0.548–0.604, a spread of 0.056 smaller than the cross-fold standard deviation within any single cell (~ 0.17 – 0.19). Background tiles decreased performance at every FP oversampling level. The $\text{fp}=1$, $\text{bgr}=0$ configuration (P2-H) was the grid winner at every metric. No data composition configuration produced a statistically distinguishable improvement over the Phase 2 reference, establishing that the limiting factor at single-patient scale was data scarcity and augmentation quality, not the ratio of existing data categories.

Phase 3. The structural pivot to 8-fold LOPO on 8 patients with on-the-fly per-epoch augmentation produced the single largest performance improvement in the project. P3-A achieved $\text{mAP50} = 0.844 \pm 0.137$ and $\text{recall} = 0.840$, a 30.3 percentage point improvement in recall over the best Phase 2 result (P2-H: 0.537). The cross-fold standard deviation simultaneously dropped from 0.194 to 0.137 despite the harder evaluation protocol, because the more biologically diverse 8-patient corpus more than compensated for the stricter patient-level holdout design.

The tinkering matrix (P3-B, 14 of 24 cells completed) identified $\text{degrees}=5$ rotation augmentation as the dominant training-time lever. The winning configuration ($\text{degrees}=5$, $\text{dfl}=1.5$, $\text{freeze}=10$, $\text{label_smoothing}=0.0$) achieved $\text{recall} = 0.866$, $\text{R_Atypisch} = 0.892$, and $\text{R_Normal} = 0.840$, gains of 5.8, 0.8, and 4.5 percentage points over P3-A. The parameter $\text{dfl}=2.0$ consistently regressed R_Normal relative to $\text{dfl}=1.5$ across all degree and freeze combinations, $\text{freeze}=0$ underperformed $\text{freeze}=10$ in every recall-relevant slice, and label smoothing was empirically inert (bit-identical fold metrics at $\text{ls}=0.0$ and $\text{ls}=0.1$). These four findings defined the production recipe ($\text{degrees}=5$, $\text{dfl}=1.5$, $\text{freeze}=10$, $\text{lr0}=0.001$, $\text{cls}=1.0$) adopted for all subsequent phases.

The automated hyperparameter tuner (P3-C, 30 iterations $\times$ 50 epochs, single P4 holdout, mAP50-95 fitness) converged on $\text{degrees}=0.0$ and $\text{dfl}=2.47$, both directions the tinkering matrix had identified as harmful to R_Normal . A single-fold tuner optimizing mAP50-95 on an Atypisch-dominant patient maximizes Atypisch performance at the expense of Normal-class generalization, confirming that the 8-fold cross-patient matrix is the stronger experimental design for this task. The production recipe from P3-B was retained.

The model-size comparison (P3-E, yolo11l, 43 M parameters) produced $\text{R_Normal} = 0.695$, the lowest value in the project and 14.5 percentage points below the P3-B winner. At approximately 170–220 training images per fold, the larger model’s randomly re-initialized classification head overfitted to the Atypisch-dominant distribution rather than learning generalizable Normal-class features. yolo11n (2.6 M parameters) was confirmed as the appropriate architecture for this corpus scale.

Phase 4. Adding patients P9–P15 (503 new images, 313 new Normal annotations) expanded the corpus to 1,564 images and 15 LOPO folds. Patient P9, contributing 195 Normal annotations, more than doubled the total Normal instance count relative to the P1–P8 corpus. P4-A (degrees=0, the production recipe with the rotation variable inadvertently unset) achieved mAP50 = 0.903, recall = 0.904, R_Atypisch = 0.913, and R_Normal = 0.890: improvements of 5.9, 9.6, 5.1, and 13.6 percentage points over the P3-A baseline at the same degrees=0 recipe setting. The scale of the R_Normal gain (+13.6 pp) from corpus expansion alone substantially exceeded the gain from the best hyperparameter intervention (+4.5 pp from rotation augmentation in P3-B).

P4-B (full production recipe with degrees=5 explicitly set) produced mAP50 = 0.887, recall = 0.895, R_Atypisch = 0.900, and R_Normal = 0.882. All P4-A vs P4-B deltas were within 1.6 percentage points, inside the cross-fold standard deviation. The rotation augmentation gain of 4–5 pp R_Normal observed at P1–P8 scale did not replicate at P1–P15 scale, consistent with the interpretation that the more diverse 15-patient corpus already provides sufficient orientation variety in the training distribution. The freeze ablation (P4-C, freeze=0) produced results within 0.4 percentage points of P4-B, confirming that backbone freezing remains appropriate at the P1–P15 scale and that the production recipe transfers to larger corpora without re-tuning. P4-B is the primary reported result for this thesis, as all training parameters are explicitly set.

Phase 5. The P5-A training run on the complete P1–P53 corpus (LOPO, up to 53 folds after empty-label filtering) was submitted to the ZHAW HPC cluster during the final preparation of this thesis. Results were not available at the time of writing and will be incorporated in the final version.

3.2. Quantitative and Qualitative Evaluation

Per-Fold Breakdown (P4-B)

Table 3 presents the complete 15-fold test results for P4-B. Each row corresponds to one patient held out from training entirely; all metrics are evaluated on that patient’s images exclusively. The last column reports the ground-truth instance counts for each class in the test set, which governs the metric sensitivity of that fold.

Table 3: P4-B per-fold test results (15-fold LOPO on P1–P15 corpus). All metrics from the held-out patient’s test split. R_Normal mean computed over n=14 folds (Fold 5 excluded; see Measurement Notes).

Fold	Patient	mAP50	Recall	R_Atypisch	R_Normal	Atyp / Normal (test GT)
1	P1	0.584	0.576	0.865	0.286	482 / 2
2	P10	0.977	0.994	1.000	0.988	5 / 20
3	P11	0.891	0.837	0.941	0.734	17 / 20
4	P12	0.934	0.931	0.861	1.000†	133 / 0
5	P13	0.901	1.000	1.000	NaN‡	0 / 7
6	P14	0.995	1.000	1.000	1.000	5 / 3
7	P15	0.954	0.936	1.000	0.872	0 / 68
8	P2	0.864	0.859	0.917	0.800	449 / 6
9	P3	0.917	0.981	0.980	0.981	60 / 4
10	P4	0.933	0.913	0.826	1.000	122 / 6
11	P5	0.661	0.631	0.333	0.929	3 / 18
12	P6	0.995	1.000	1.000	1.000	0 / 6
13	P7	0.995	1.000	1.000	1.000	8 / 12
14	P8	0.880	0.912	1.000	0.824	0 / 24
15	P9	0.818	0.850	0.769	0.930	4 / 195
Mean	—	0.887	0.895	0.900	0.882	—

†Fold 4 (P12): P12 has 133 Atypisch instances and 0 Normal ground truth; R_Normal = 1.000 is vacuously true. ‡Fold 5 (P13): per-class positional indexing measurement error; true R_Normal = 1.000 confirmed from confusion matrix output. Excluded from R_Normal mean.

Per-Class Recall Analysis

The per-class recall gap between R_Atypisch and R_Normal narrowed from approximately 9 percentage points at the Phase 3 baseline (P3-A: R_Atypisch = 0.884, R_Normal = 0.795, computed from the P3-B tinkering matrix default cell) to 1.8 percentage points at the Phase 4 best run (P4-B: R_Atypisch = 0.900, R_Normal = 0.882). The narrowing reflects two simultaneous effects: a 12.9 percentage point absolute improvement in R_Normal driven primarily by patient P9’s 195 Normal annotations, and a 1.6 percentage point improvement in R_Atypisch from the additional Atypisch-dominant patients P10–P15. Neither effect alone explains the gap reduction; the combination of more Normal training signal and better cross-patient Atypisch diversity together narrowed the difference.

R_Normal is the more clinically constrained metric throughout the project. The R_Normal sequence across key milestones traces the cumulative contribution of each methodological intervention: 0.695 (P3-E, yolo11l), 0.795 (P3-A, on-the-fly augmentation and LOPO), 0.840 (P3-B, rotation augmentation), 0.890 (P4-A, patient P9 added), 0.882 (P4-B). The single largest R_Normal

gain at any step (+9.5 pp from P3-B to P4-A) came from corpus expansion, not from hyperparameter adjustment, which is consistent with the Phase 2 DoE’s controlled negative result on data composition within a fixed corpus. Improving rare-class recall in this domain requires more biologically diverse patients rather than a better arrangement of existing data.

Fold-Level Structural Patterns

Two fold-level patterns repeat across all Phase 3 and Phase 4 runs and reflect properties of the corpus rather than model behaviour.

Fold 1 (P1 holdout) is the persistent weakest fold. P4-B Fold 1 produced $\text{mAP50} = 0.584$, $\text{recall} = 0.576$, and $\text{R_Normal} = 0.286$. P1’s test set contains only 2 Normal instances; one missed Normal cell changes R_Normal by 50 percentage points. The fold is also affected by the annotation regime gap: P1 was annotated under the original `Pos_neg` directory structure prior to CVAT-assisted model-assisted labelling, which produces a different spatial distribution of bounding boxes relative to P2–P15. The Fold 1 weakness is consistent across all Phase 3 and Phase 4 runs and is not addressable by adjusting training hyperparameters; it reflects the structural distance between P1’s annotation regime and the rest of the corpus.

Fold 11 (P5 holdout) is the second-weakest fold. P5 contributes only 3 Atypisch instances to the test set; each missed Atypisch cell changes R_Atypisch by 33 percentage points. P4-B Fold 11 produced $\text{R_Atypisch} = 0.333$ (2 of 3 detected). This is a measurement sensitivity effect, not a model regression: P5’s Normal recall (0.929) was among the highest in the fold table, indicating that the model handles Normal-dominant distributions correctly when trained without P5. The same structural fragility applies to Fold 15 (P9 holdout: 4 Atypisch, $\text{R_Atypisch} = 0.769$) and Fold 10 (P4 holdout: 6 Atypisch, $\text{R_Atypisch} = 0.826$), where one or two missed cells produce double-digit metric swings. In all these cases, per-fold recall values are informative as upper and lower bounds rather than as precise point estimates.

Measurement Notes

A per-class recall extraction issue affected all LOPO runs. The utility function indexed Ultralytics’ per-class result arrays by position rather than by the `ap_class_index` field. When a holdout patient contained only one class and the model produced predictions for that class only, Ultralytics returned a one-element rather than a two-element result array, causing positional indexing to assign the single element to R_Atypisch and report R_Normal as NaN. This affected Fold 5 (P13 holdout) in every run; the true R_Normal for that fold is 1.000, confirmed from the per-fold confusion matrix. Affected folds are excluded from per-class cross-fold means throughout this thesis and are clearly marked in all fold tables. The aggregate metrics mAP50 and overall Recall are computed internally by Ultralytics and are not affected by this issue.

3.3. Evaluation Visualization

Training convergence for P4-B is shown in Figure 9 for a representative fold (Fold 8, P2 holdout: $\text{mAP50} = 0.864$, $\text{recall} = 0.859$). The training and validation loss curves converge within approximately 150 epochs on the majority of folds, with YOLO’s early-stopping criterion (patience = 100) triggering at the validation mAP50 plateau. The convergence pattern is representative of all folds

except Fold 1 (P1 holdout), where the annotation regime gap described above leads to a longer plateau and lower final validation mAP50 before early stopping.

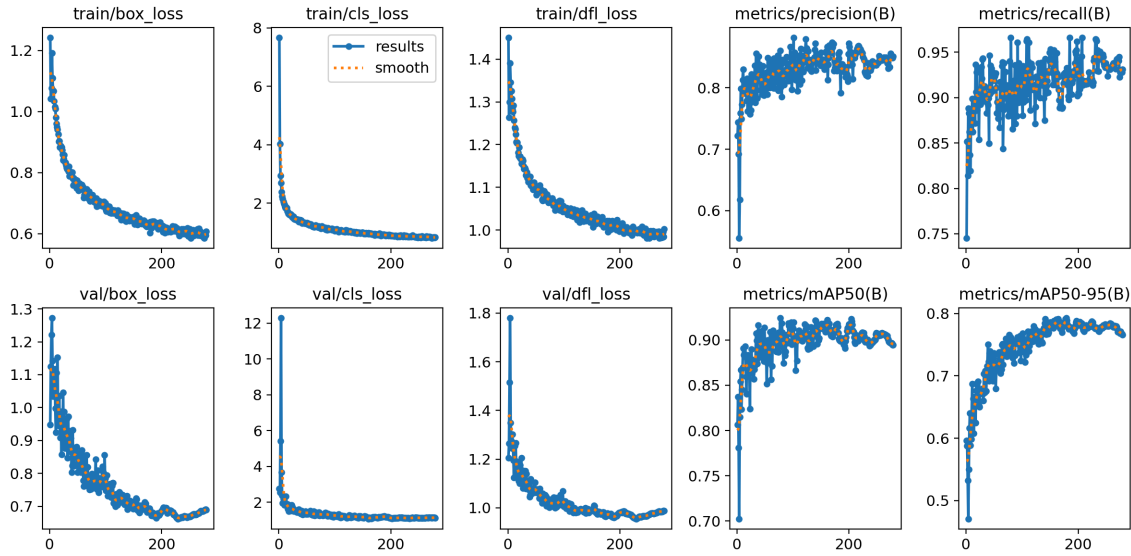


Figure 9: Training convergence for P4-B Fold 8 (P2 holdout). Panels from top left: box loss, classification loss, and distribution focal loss for training and validation splits; precision, recall, mAP50, and mAP50-95 across epochs. Convergence is representative of all non-structurally-weak folds.

The normalized confusion matrix for the same fold is shown in Figure 10. Rows correspond to ground-truth classes and columns to predicted classes; values are row-normalized to recall fractions. The matrix reflects the class asymmetry that persists throughout the project: $R_{\text{Atypisch}} = 0.917$ (437 of 449 Atypisch instances detected) against $R_{\text{Normal}} = 0.800$ (5 of 6 Normal instances detected in this fold). The background column captures false negatives: instances the model assigned to no class at the evaluation confidence threshold of 0.58.

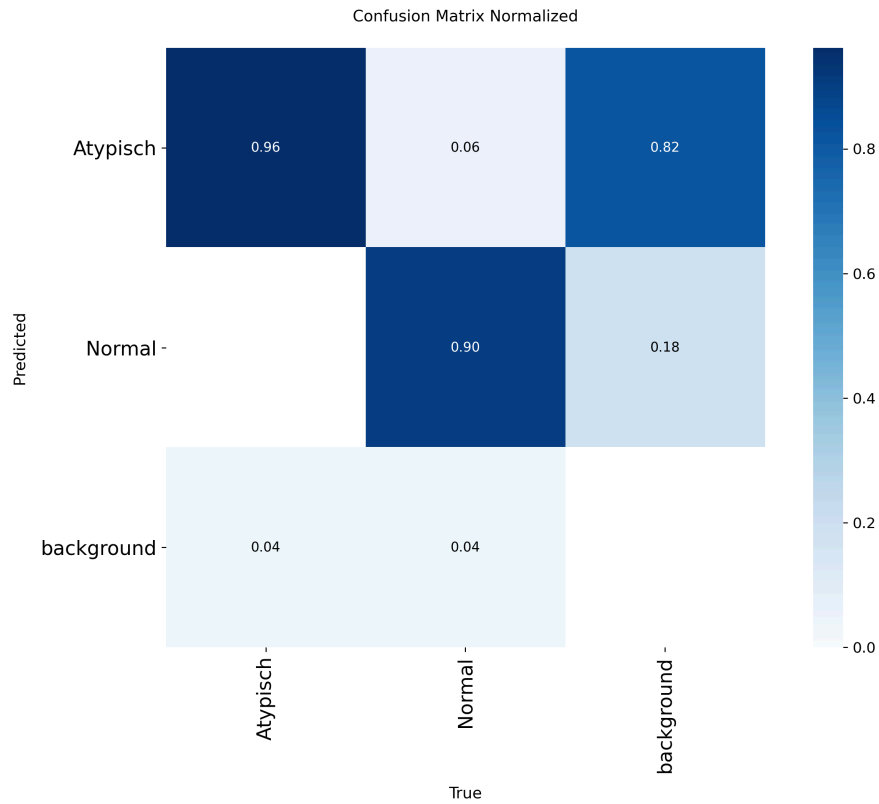


Figure 10: Normalized confusion matrix for P4-B Fold 8 (P2 holdout). Rows are ground-truth classes; columns are predicted classes. Values are row-normalized recall fractions. Background represents tiles with no detection at the evaluation threshold.

The precision-recall curve for the same fold is shown in Figure 11. Both classes maintain high precision across a broad recall range before the precision cliff, reflecting the model's discrimination capacity at sub-deployment confidence levels. The Atypisch curve sits above the Normal curve at all recall values, consistent with the majority-class training advantage. At the deployed threshold of 0.58, both curves are in their high-precision operating region.

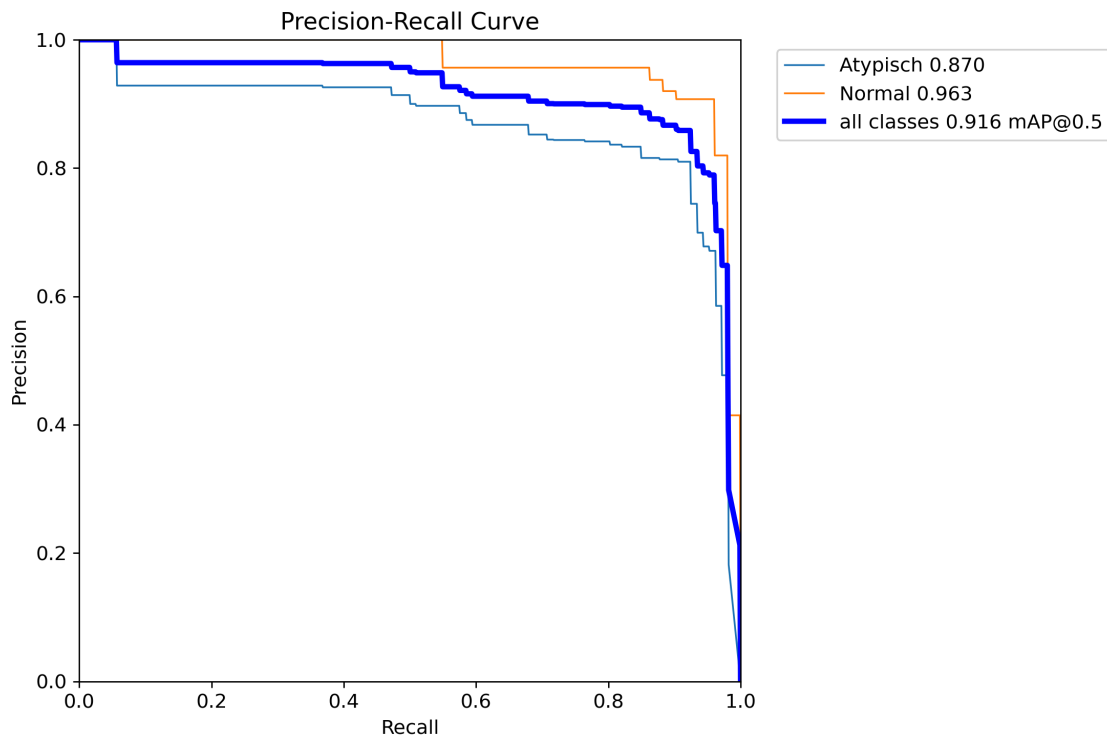


Figure 11: Box precision-recall curves for P4-B Fold 8 at IoU threshold 0.50. Atypisch (blue) and Normal (orange) curves illustrate the per-class discrimination capacity across confidence thresholds.

The morphological appearance of mast cells in aspirate tiles is illustrated in Figure 12. Cells appear at diverse orientations and staining intensities with no canonical vertical alignment. The 5-degree rotation augmentation identified as the dominant training-time lever in P3-B directly targets this orientation variability: the improvement in R_{Normal} of 4.5 percentage points from rotation augmentation at P1–P8 scale is consistent with a model that learns orientation-invariant cell representations rather than exploiting a spurious vertical bias in the training data.

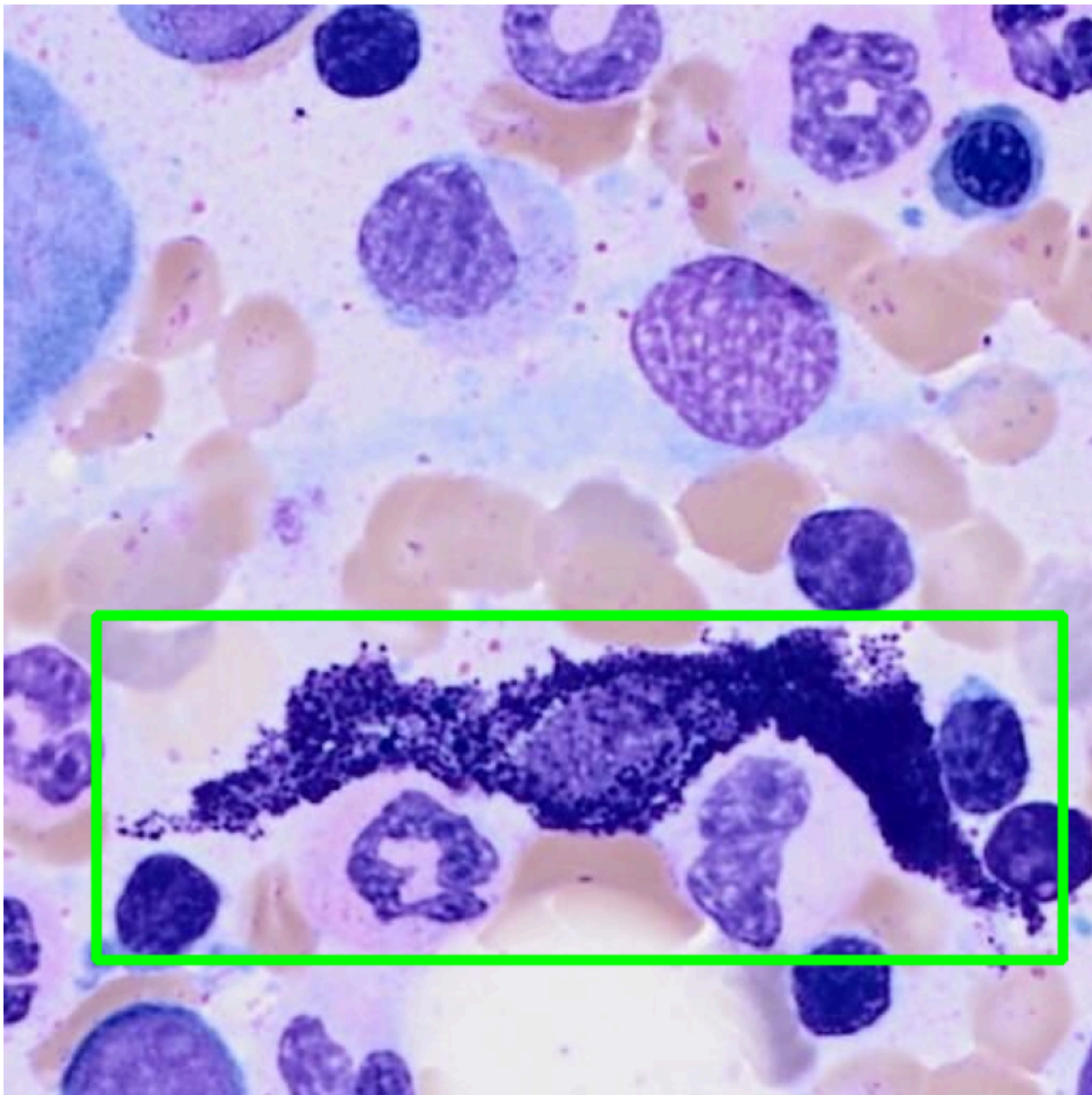


Figure 12: Sample bone marrow aspirate tile showing a mast cell with the characteristic spindle-shaped cytoplasmic morphology that distinguishes atypical from normal cells under microscopy.

3.4. Real World Usage

Tests With Unseen Data

After Phase 1 training, an inference run was conducted on the complete first-patient slide set, approximately 20,000 tiles, with all training images excluded by filename. This was an exploratory step: the model had only seen a fraction of the slide during training, and running it across the full scan gave an early read on its behavior at deployment scale. The resulting confidence distribution, shown in Figure 13, illustrates how the Phase 1 model separated high-confidence detections from low-confidence responses across the full tile set. The distribution was reviewed together with Robin Kosche and Dr. Stefan Glüge to understand where the model was uncertain and to guide the direction of subsequent annotation and training work.

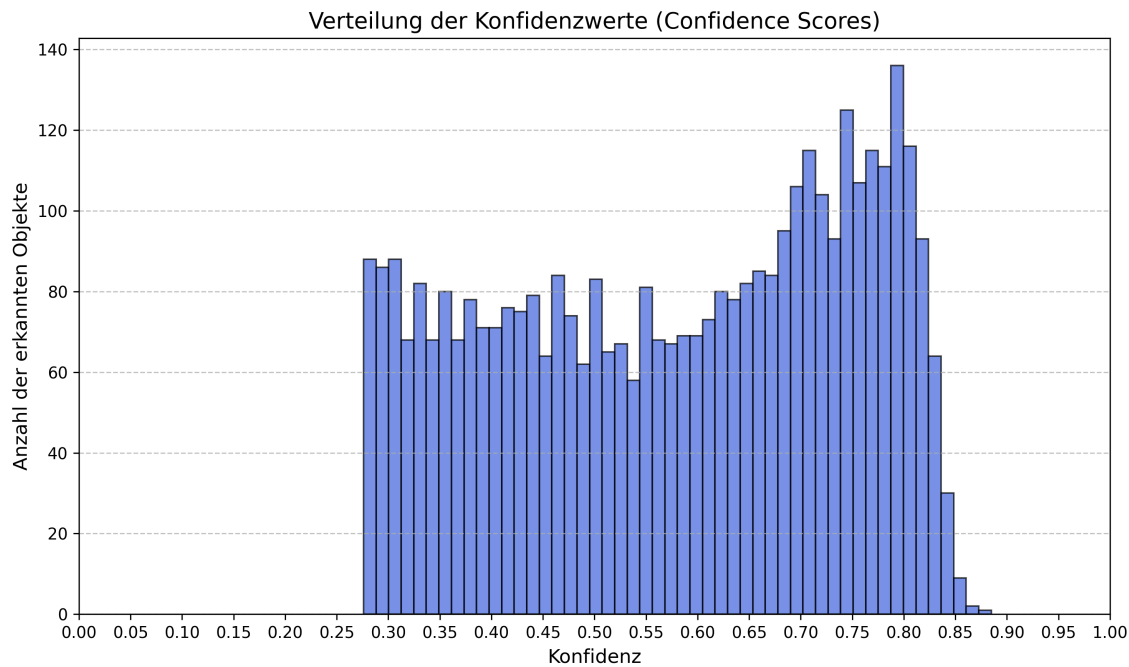


Figure 13: Confidence distribution of the Phase 1 model across the full P1 slide set (~20,000 tiles). The distribution reflects model behavior before any threshold was fixed.

The confidence distribution in Figure 13 shows a clear separation between a high-confidence detection peak and a broad low-confidence region. The deployment threshold of 0.58 was calibrated from this distribution: it sits above the low-confidence bulk while retaining the high-confidence peak, capturing the detections the model assigns the most discriminative evidence to while discarding uncertain responses. At this threshold, the expected clinical behaviour on confirmed SM patients is a high Atypisch fraction with few false alarms in the majority of tiles, requiring manual review primarily for tiles near the threshold boundary rather than for systematic class errors.

The MastCellDetector desktop application was delivered to USZ alongside the P3-D model weights as the primary clinical tool for patient-scale inference. The application runs inference on complete patient tile folders, streaming YOLO label files to disk in real time, and presents results through the gallery and statistics interface described in the Methodology. The gallery filter tabs (All, With detections, No detection, Low confidence, Atypisch only, Normal only) enable a haematologist to navigate directly to tiles most likely to require manual review, avoiding the need to scroll through tens of thousands of tiles per patient. The Statistics tab produces the patient-level Atypisch fraction summary relative to the WHO 25% diagnostic threshold, providing the clinical decision support output that motivated the project.

4. Discussion

4.1. Interpretation of Results

4.2. Clinical Relevance

4.3. Error Analysis

5. Conclusion

5.1. Summary of Key Findings

5.2. Limitations and Challenges

5.3. Future Research Directions

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, and optimization of academic reading flow. Specifically, the AI models (Gemini 2.5 Pro, Claude Sonnet 4.5, and Claude Sonnet 4.6) were employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

- [1] T. Ghete *et al.*, “Models for the marrow: A comprehensive review of AI-based cell classification methods and malignancy detection in bone marrow aspirate smears,” *HemaSphere*, vol. 8, no. 12, p. e70048, 2024, doi: <https://doi.org/10.1002/hem3.70048>.
- [2] H. Sazak and M. Kotan, “Automated blood cell detection and classification in microscopic images using YOLOv11 and optimized weights,” *Diagnostics*, vol. 15, no. 1, 2025, doi: [10.3390/diagnostics15010022](https://doi.org/10.3390/diagnostics15010022).
- [3] J. Diaz, A. Deb, A. Lyubimova, L. Santos-Carrion, C. Romagnoli, and J. Barrientos, “Machine learning-based detection model for leukemia cells in peripheral blood smears using YOLOv11-large,” *Blood*, vol. 146, p. 6123, 2025, doi: <https://doi.org/10.1182/blood-2025-6123>.
- [4] G. Jocher, J. Qiu, and A. Chaurasia, *Ultralytics YOLO*. (Jan. 2023). Available: <https://github.com/ultralytics/ultralytics>
- [5] F. Vollmer, “MastCellDetector: Desktop application for mast cell detection and classification.” 2026. Available: https://github.com/kekslol24/desktop_app
- [9] Gemini 2.5 Pro. (n.d.). Google Gemini. Retrieved October 1, 2025, from <https://gemini.google.com/app>
- [10] DeepL Translate: The world’s most accurate translator. (n.d.). <https://www.deepl.com/>

8. Lists

List of Figures

Figure 1	Analysis tab of the Gradio prototype showing uploaded image tiles with YOLO detection overlays.	18
Figure 2	Research tab showing ground-truth annotations alongside model predictions for direct visual comparison.	18
Figure 3	Research tab metrics output: precision-recall curves and performance plots generated from the ground-truth evaluation run.	19
Figure 4	Statistics dashboard of the Gradio prototype showing per-class detection counts and the Atypisch fraction relative to the WHO 25% diagnostic threshold.	19
Figure 5	Gallery view of the desktop application showing all image tiles for a patient folder with sidebar controls for inference configuration.	20
Figure 6	Gallery filtered to tiles with detections, allowing rapid navigation to images where the model found mast cells.	21
Figure 7	Annotation editor with interactive bounding boxes; boxes can be added, moved, resized, or deleted directly in the application without external tooling.	22
Figure 8	Statistics tab aggregating per-image detections into a patient-level summary with the Atypisch fraction displayed against the WHO 25% diagnostic threshold.	23
Figure 9	Training convergence for P4-B Fold 8 (P2 holdout). Panels from top left: box loss, classification loss, and distribution focal loss for training and validation splits; precision, recall, mAP50, and mAP50-95 across epochs. Convergence is representative of all non-structurally-weak folds.	30
Figure 10	Normalized confusion matrix for P4-B Fold 8 (P2 holdout). Rows are ground-truth classes; columns are predicted classes. Values are row-normalized recall fractions. Background represents tiles with no detection at the evaluation threshold.	31
Figure 11	Box precision-recall curves for P4-B Fold 8 at IoU threshold 0.50. Atypisch (blue) and Normal (orange) curves illustrate the per-class discrimination capacity across confidence thresholds.	32
Figure 12	Sample bone marrow aspirate tile showing a mast cell with the characteristic spindle-shaped cytoplasmic morphology that distinguishes atypical from normal cells under microscopy.	33

Figure 13 Confidence distribution of the Phase 1 model across the full P1 slide set (~20,000 tiles). The distribution reflects model behavior before any threshold was fixed. . . . 34

List of Tables

Table 1	Full corpus summary across P1-P53.	10
Table 2	Milestone run progression across all experimental phases. Phase 1–2 values are from stratified k-fold validation splits; Phase 3–4 values are from LOPO test splits. ★ = production recipe confirmed by P3-B tinkering matrix. ★★ = best complete run, cited as the primary performance reference throughout this thesis.	25
Table 3	P4-B per-fold test results (15-fold LOPO on P1–P15 corpus). All metrics from the held-out patient’s test split. R_Normal mean computed over n=14 folds (Fold 5 excluded; see Measurement Notes).	28

9. Attachments